Aishwarya Mishra

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Career Objective

Final year Computer Science (AIML) student with a strong foundation in Machine Learning, Deep Learning, Natural Language Processing, and Generative AI. Experienced in building GenAI/RAG pipelines, predictive ML models, with working knowledge of System Design, SQL, and data analytics tools like Tableau. Proficient in Python and Java with strong problem-solving skills and a keen interest in applying AI to real-world engineering problems.

Education

Degree	Institute	Year	Percentage/CGPA
B.Tech, Computer Science (AIML)	Pranveer Singh Institute of Technology, Kanpur	2023–2027	9.2 (upto 6th sem)
Intermediate	Mercy Memorial School, Kanpur	2022–23	92%
High School	Mercy Memorial School, Kanpur	2020–21	97%

Projects

ClinRAG – MIMIC IV Research Assistant

Tech Stack: GenAI, RAG, LangChain, BeautifulSoup4, Groq API, Streamlit

- Engineered a Retrieval-Augmented Generation (RAG) pipeline over the Physionet MIMIC-IV dataset, supporting dual General and Clinical research modes.
- Architected core components including a conversational chatbot, website summarization module, and prompt engineering workflows to improve response relevance.
- Integrated LangChain, BeautifulSoup4, and the Groq API within a Streamlit interface to deliver an end-to-end research assistant.
- Under review to be published as a Research Paper under IEEE Conference 2026.

Network Security API

Tech Stack: RNN, FastAPI, uvicorn, traditional ML models, scikit-learn, pandas, numpy

- Developed a Recurrent Neural Network (RNN) for binary classification of malicious URLs using the Kaggle Phishing Dataset and compared the performance with traditional ML models to choose the best model for performing classification.
- Designed and deployed a RESTful API with FastAPI and uvicorn to serve real-time phishing detection predictions.
- Performed data preprocessing, feature engineering, and model evaluation using scikit-learn, pandas, and numpy.

Next Word Predictor

Tech Stack: LSTM, NLP, TensorFlow, NLTK

- Designed and trained an LSTM-based Recurrent Neural Network for sequence modeling and next-word text generation.
- Built an NLP preprocessing pipeline – tokenization, sequence padding, and corpus cleaning – on Shakespeare's Hamlet dataset using NLTK.
- Implemented model training and evaluation in TensorFlow, with performance visualization using Pandas and Matplotlib.

Technical Skills

Core: Machine Learning, Deep Learning, Natural Language Processing, Generative AI, Agentic AI.
Languages & Web: Python, Java, SQL, HTML, CSS.
Databases: MongoDB, Structured & Semi-structured Data.
Tools: VS Code, GitHub Actions, Postman, Tableau.
Others: Problem Solving, Soft Skills.

Subjects

Database Management System, Computer Networks.

Certifications & Achievements

- Complete ML, DL, and NLP – Udemy (certificate).
- Machine Learning 1 – Columbia University.
- Complete Generative AI with LangChain and Huggingface – Udemy (certificate).
- Data Analytics using Power BI, Tableau, SQL – Udemy (certificate).
- Won Inter-School Declamation Competitions.
- Have been on the position of Head-Girl of the school.
- NCC B certification on the post of senior under-officer.
- Solved 500+ LeetCode problems.